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EEE 4709-Artificial Intelligence & Machine Learning

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Road Condition Monitoring System Using AI: A Novel Approach

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WEBSITE LINK

<https://roadconditiondetectorwebapp.streamlit.app/>

Brief Description

This research presents an automated system that uses artificial intelligence to monitor road conditions, **specifically detecting potholes and water accumulation during rainy seasons**. The system employs a **Convolutional Neural Network (CNN)** to analyze aerial images captured by surveillance drones, enabling cities to better plan and prioritize road maintenance activities. The AI model processes these images to classify roads as either in good condition or requiring maintenance, allowing for more efficient allocation of resources.

Training was conducted on a dataset of 4,200 road images, with validation performed on 900 additional images. The CNN architecture follows a VGG-like structure with **five convolutional blocks** of increasing filter sizes (32→64→128→256→512), each followed by max pooling operations. The network includes dropout layers to prevent overfitting and uses binary cross-entropy as the loss function. By optimizing the decision threshold to 0.3 instead of the conventional 0.5, the model **achieved classification accuracy exceeding 95%**, representing a significant improvement over comparable study.

Unlike most existing approaches that focus solely on structural defects, this system successfully identifies both potholes and water accumulation issues, providing a more comprehensive assessment of road maintenance needs. According to research cited, poor road conditions cost billions annually in vehicle repairs and operating costs, highlighting the economic importance of timely maintenance.

The proposed solution addresses several limitations of traditional inspection methods: subjective assessment, inefficient resource allocation, delayed response times, and limited coverage. The integration of GPS-tagged imagery with AI analysis creates a practical framework for municipal authorities to implement proactive maintenance strategies. The system is designed to operate with images collected by drone surveillance systems that capture aerial footage of city roads, with the data sent to a central server for processing.

MODEL Results:



